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Heavy metal pollution: the Minimata case

Mercury (chemical symbol Hg) is a 'heavy metal', one of a set of metal elements that are toxic. Mercury has been used for thermometers, barometers, electrical switches, lighting, batteries and in the manufacture of sodium hydroxide. The metal itself is moderately toxic, but becomes severely toxic when it forms compounds with carbon.

Early last century, the people of the Japanese fishing village of Kumamoto, about 600 km from Tokyo, hoped to expand their quiet little fishing village to a larger, more prosperous town. Consequently they welcomed the arrival of the chemical producing company, Chisso Corporation. The Chisso plant produced, among other forms of waste, mercury compounds that were freely dumped into Minimata Bay. The villagers depended on this bay for fish. Almost immediately the deadly effects the chemicals had on the fish became evident. The company recognised the damage and paid the villagers for permission to ruin their fishing industry. The dumping of mercury compounds continued from the late 1920s up until 1968. Minimata Bay flowed out to the Shiranui Sea (an inland sea), poisoning the aquatic species along its path.

By the 1950s rumours began to spread that the village was haunted by a spirit that took over the bodies of Kumamoto residents. Many cats, and some dogs, began to do 'weird dances' on the streets before dropping dead. Children were also behaving strangely and soon adults followed. People began to develop speech and vision problems, others began screaming wildly, muscle spasms led to paralysis and suicides occurred. The bizarre condition became known as *'Minimata Disease'*.

People then suspected some form of infectious disease was attacking the central nervous system. The corporation's doctor, Dr Hajim Hosokawa, discovered the connection between a fish diet and the condition. This led him to analyse the fish and identify the heavy mercury contamination.

1 Chisso Corporation offered compensation pay-offs in exchange for affected people signing waivers that surrendered their rights to court action. Give reasons why the company would be willing to pay the villagers to destroy their fishing industry.

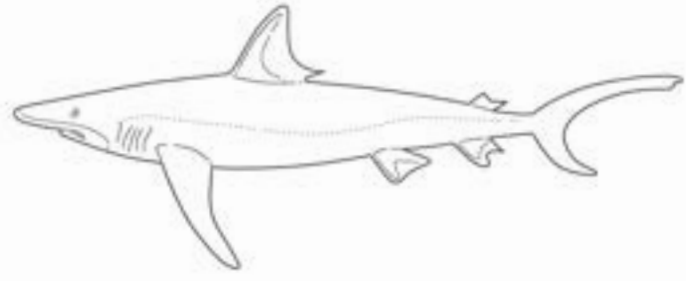
2 Suggest why children and animals were the first to be affected by the toxin.

3 Would the effects on the villagers have been the same if the Shiranui Sea not been an inland sea? Explain your answer.

4 Give two reasons why local residents might welcome the arrival of a chemical company.

5 Name two uses for mercury.

A television current affairs program ran a 'special' on the effects of mercury on pregnant women. One piece of advice from a researcher was to advise pregnant women not to eat fish such as swordfish or sharks.



6 Why would the researcher specifically mention these two types of fish?

Research

7 The Mad Hatter in *'Alice's Adventures in Wonderland'* (Lewis Carroll, 1865) shows classic signs of mercury poisoning. Find out how the phrase *'Mad as a hatter'* originated.

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